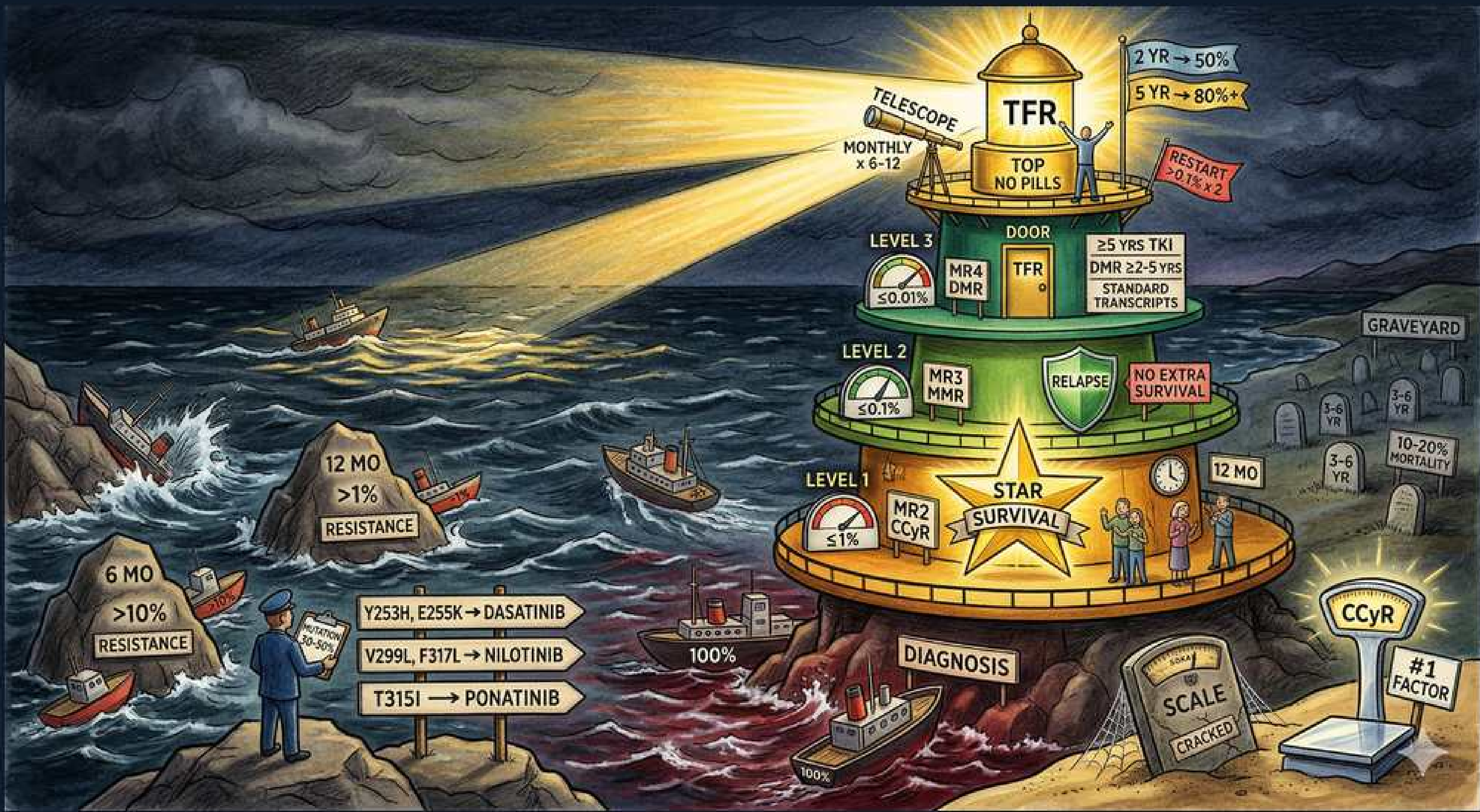
CML — Response Monitoring, Resistance & Treatment-Free Remission



1. Diagnosis 100% IS

WHAT YOU SEE
Bottom of lighthouse tower, DIAGNOSIS, 100%.

WHAT IT ENCODES
At diagnosis BCR::ABL1 is set at 100% on the International Scale. Quantitative RT-PCR every ~3 months tracks log reduction; failure to hit milestones triggers mutation testing and a switch.

BOARD TRAP
Baseline 100% is a reference point on the IS — response is judged by log-reduction milestones at fixed time points, not by symptoms.

2. 6 mo >10% = resistance

WHAT YOU SEE
Sign on the rocks: 6 MO >10% RESISTANCE.

WHAT IT ENCODES
BCR::ABL1 >10% IS at 6 months indicates an inadequate (warning/failure) response. After confirming adherence and drug interactions, check kinase-domain mutations and consider switching TKI.

BOARD TRAP
Before calling true resistance, exclude non-adherence and CYP3A4 drug interactions — pseudo-resistance is the commonest cause of a rising transcript.

3. 12 mo >1% = resistance

WHAT YOU SEE
Sign: 12 MO >1% RESISTANCE.

WHAT IT ENCODES
By 12 months the target is CCyR (≤1% IS / MR2). Remaining >1% at 12 months is treatment failure warranting mutation analysis and a change of therapy.

BOARD TRAP
≤1% IS (CCyR) at 12 months is the key 'on-track' milestone — not MMR; MMR (≤0.1%) is desirable but not mandatory at 12 months.

4. Mutation-directed switching

WHAT YOU SEE
Signpost: Y253H/E255K→dasatinib; V299L/F317L→nilotinib; T315I→ponatinib.

WHAT IT ENCODES
Kinase-domain mutations guide TKI choice: Y253H/E255K/F359 favor dasatinib; V299L/F317L favor nilotinib; T315I requires ponatinib (or asciminib/omacetaxine).

BOARD TRAP
V299L and F317L are resistant to DASATINIB — so they are switched to nilotinib; T315I is resistant to BOTH 2nd-gen drugs and needs ponatinib. Match the mutation to the drug, do not just escalate blindly.

5. MR2 / CCyR (≤1%) Level 1

WHAT YOU SEE
LEVEL 1 platform, MR2 CCyR, ≤1%, STAR SURVIVAL.

WHAT IT ENCODES
MR2/CCyR (≤1% IS) is the survival milestone — patients reaching it have overall survival approaching normal. This is the floor of an acceptable response.

BOARD TRAP
CCyR drives survival; deeper responses (MMR, MR4) add the option of TFR but do not dramatically further improve survival.

6. MR3 / MMR (≤0.1%) Level 2

WHAT YOU SEE
LEVEL 2, MR3 MMR, ≤0.1%, NO EXTRA SURVIVAL, RELAPSE.

WHAT IT ENCODES
MR3 = major molecular response (≤0.1% IS, 3-log reduction). It confers excellent prognosis and low relapse risk but does not add survival over CCyR; it is the gateway toward deeper responses.

BOARD TRAP
MMR alone is NOT deep/durable enough to stop the TKI — attempting TFR at MR3 risks molecular relapse.

7. MR4 / DMR Level 3

WHAT YOU SEE
LEVEL 3, MR4 DMR, ≥2 yrs TKI, DMR ≥2 yrs, STANDARD TRANSCRIPTS.

WHAT IT ENCODES
Deep molecular response (MR4 ≤0.01% / MR4.5 ≤0.0032%) sustained for ≥2 years, on a TKI for ≥3–5 years total, with a standard quantifiable (p210) transcript, is the prerequisite to attempt treatment-free remission.

BOARD TRAP
TFR eligibility requires a STANDARD transcript that can be tracked sensitively — atypical transcripts can't be monitored deeply enough to safely stop.

8. TFR (stop therapy)

WHAT YOU SEE
Lighthouse top, TFR, NO PILLS, monthly x6–12, RESTART ≥MR3, 2 yr 50% / 5 yr 80%.

WHAT IT ENCODES
In treatment-free remission ~50% maintain remission; relapses cluster in the first 6 months, so PCR is checked monthly x6–12. Loss of MMR (>0.1%) triggers prompt TKI restart, which re-induces response in nearly all.

BOARD TRAP
Restart the TKI when MMR is lost (>0.1% / loss of MR3), NOT when survival or symptoms change — molecular relapse is the stopping rule, and it is reversible.

9. Sokal/ELTS risk #1 factor

WHAT YOU SEE

Cracked SCALE, CCyR, #1 FACTOR.

WHAT IT ENCODES

Baseline risk scores (Sokal, EUTOS Long-Term Survival/ELTS) stratify patients; achieving CCyR/deep response is the strongest determinant of long-term outcome. High-risk scores favor potent 2nd-gen TKIs upfront.

BOARD TRAP

Depth of molecular response achieved on therapy outweighs baseline risk score for ultimate prognosis — a high-Sokal patient who reaches MR4 does well.

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10. Graveyard 10–20% mortality

WHAT YOU SEE

GRAVEYARD with tombstones, 10–20% MORTALITY.

WHAT IT ENCODES

Despite TKIs, ~10–20% still die from CML or progression over time, mostly those who fail to achieve milestones, progress to blast phase, or harbor T315I/complex karyotypes.

BOARD TRAP

TKIs transformed CML into a chronic disease but did not make it universally curable — non-responders and blast-phase patients still have substantial mortality.
